# Cap LoRa868

SKU:U201































## Description

**Cap LoRa868** is a high-performance LoRa communication and GNSS global navigation expansion module designed for **Cardputer-Adv** / **CardputerZero**. The LoRa module is based on the SX1262 chip solution, equipped by default with an **external SMA antenna**, providing stronger anti-interference capability and long-distance communication performance. It supports the **868MHz ~ 923MHz frequency band** and offers long-range, low-power wireless links for IoT applications.

The GPS module adopts the AT6668-based solution, with a built-in ceramic antenna for precise and reliable satellite positioning services. It supports **multi-frequency, multi-mode GNSS signal** reception, enabling the reception of signals from multiple satellite navigation systems (including GPS, BD2, BD3, GLONASS, GALILEO, and QZSS, etc.) and supports both multi-system joint positioning and standalone positioning with strong anti-interference capability and high positioning accuracy. Even in weak signal environments, it can quickly obtain high-accuracy location data and is suitable for remote data collection, smart cities, smart homes, vehicle positioning and navigation, and IoT device positioning.

## Features

- Integrated LoRa modulation technology and GNSS global positioning system
- Operating frequency 868 ~ 923 MHz
- Supports FSK, GFSK, MSK, GMSK, LoRa, OOK modulation modes
- LoRa receive sensitivity: as low as -147 dBm (in LoRa low data rate mode)
- LoRa maximum output power: +20 dBm
- LoRa external SMA antenna interface, GPS built-in ceramic antenna
- Positioning supports multiple satellite navigation systems (GPS/QZSS/BD2/BD3/GAL/GLO)
- Multi-frequency multi-system reception
- Multi-channel
- Development Platform:
  - Arduino
  - UiFlow2

## Includes

- 1 x Cap LoRa868
- 1 x LoRa Rubber Duck Antenna

## Applications

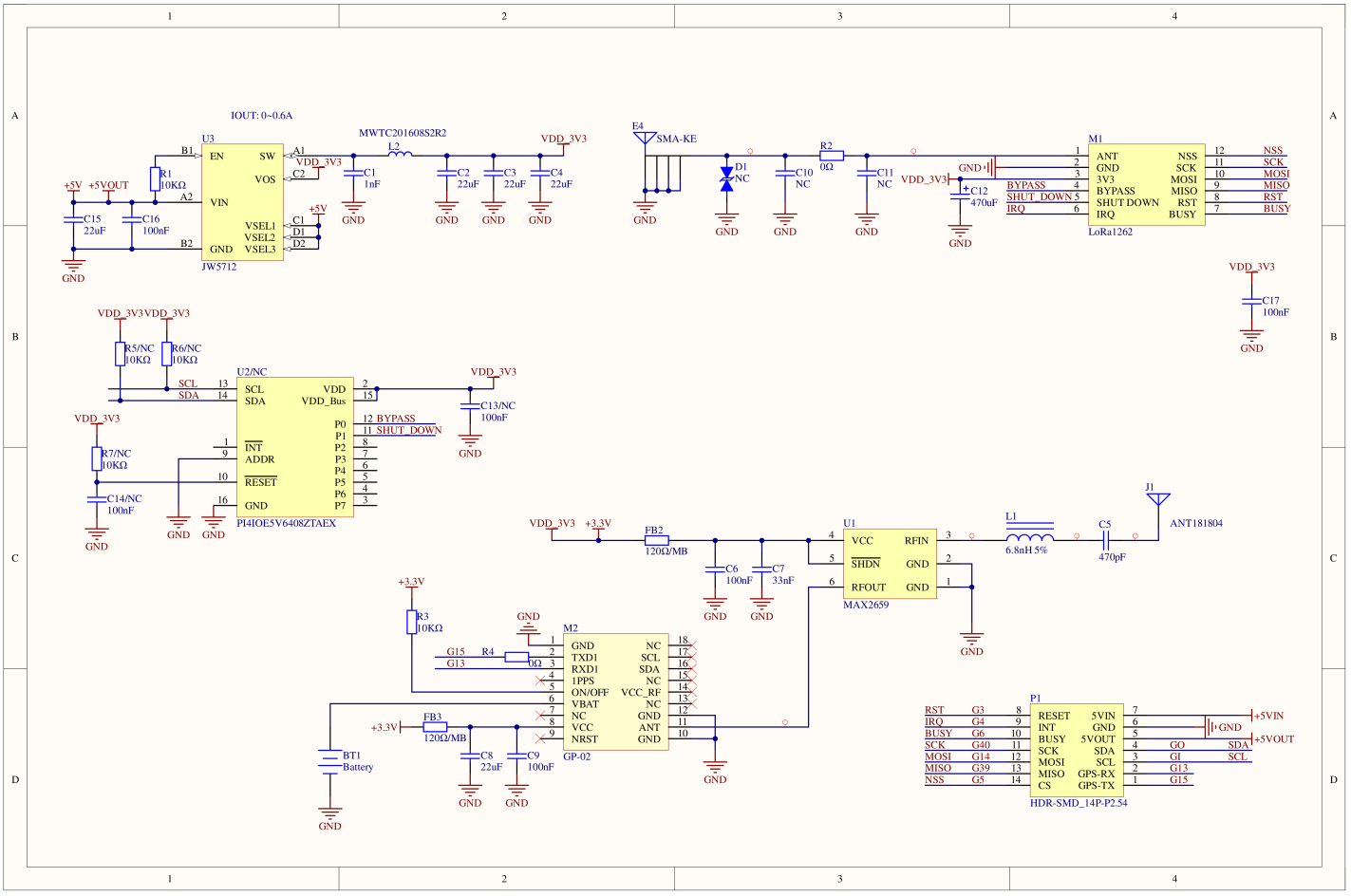
- Remote data collection
- Smart cities
- Smart homes
- Vehicle positioning and navigation
- IoT device positioning

# Specifications

| Specification | Parameter |
| --- | --- |
| LoRa Module | Stamp LoRa-1262@SX1262 |
| LoRa Frequency | 868 ~ 923 MHz |
| LoRa Bitrate | Up to 300kbps |
| LoRa Interface | SPI |
| LoRa Antenna | Rubber Duck Antenna @3dBi Length: 108 x 9.3mm |
| LoRa TX Power | +20 dBm |
| LoRa Sensitivity | -147 dBm (LoRa low data rate mode) |
| Modulation Modes | FSK, GFSK, MSK, LoRa, OOK |
| GPS Module | ATGM336H-6N@AT6668 |
| GPS Interface | UART, default 115200bps@8N1 |
| Supported GNSS | GPS/QZSS/BD2/BD3/GAL/GLO |
| Frequency Points | BDS: B1I+B1C<br>GPS/QZSS/SBAS: L1<br>GALILEO: E1<br>GLONASS: R1 |
| GPS Channels | 50 channels |
| GPS Accuracy | <1.5m (CEP50) |
| GPS Update Rate | Max 10Hz |
| GPS Protocol | NMEA 0183 4.1 |
| GPS Sensitivity | Tracking: -162dBm<br>Acquisition: -160dBm<br>Cold Start: -148dBm |
| GPS TTFF | Cold Start: 23s<br>Hot Start: 1s |
| Working Current | GPS on, LoRa standby: DC 5V@29.24mA<br>GPS sleep, LoRa standby: DC 5V@3.16mA<br>GPS sleep, LoRa TX max power: DC 5V@129.72mA<br>GPS on, LoRa TX max power: DC 5V@155.03mA |
| Product Size | 84.0 x 24.0 x 15.2mm |
| Package Size | 170.0 x 120.0 x 17.0mm |
| Product Weight | 30.0g |
| Gross Weight | 32.2g |

# Schematics

- Cap LoRa868 Schematics PDF




# PinMap

## Cap-Bus

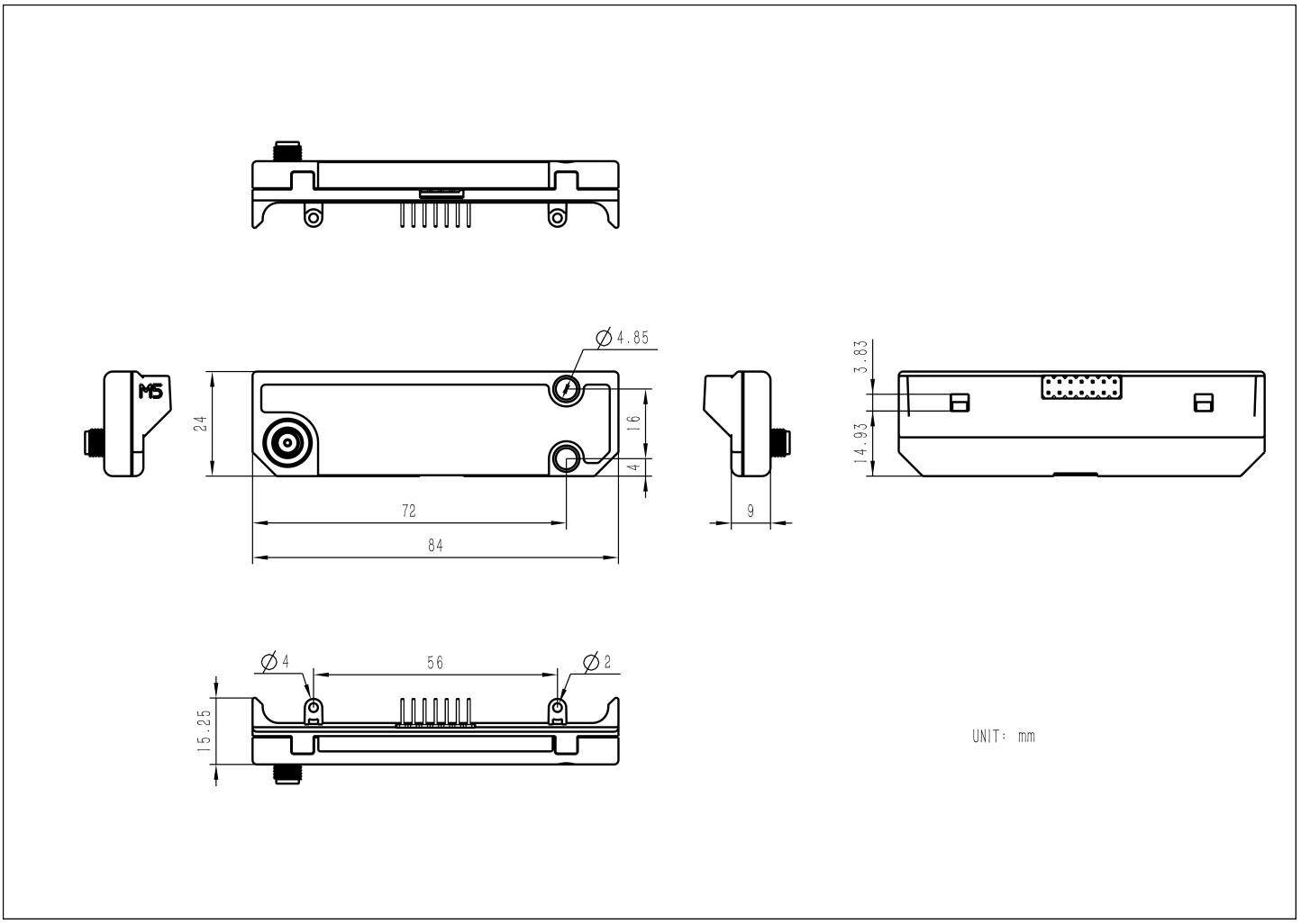
| PIN | LEFT | RIGHT | PIN |
|---|---|---|---|
| GPS_TX | 1 | 14 | LoRa_NSS |
| GPS_RX | 2 | 13 | LoRa_MISO |
| SCL | 3 | 12 | LoRa_MOSI |
| SDA | 4 | 11 | LoRa_SCK |
| 5V_OUT | 5 | 10 | LoRa_BUSY |
| GND | 6 | 9 | LoRa_IRQ |
| 5V_IN | 7 | 8 | LoRa_RST |

| Cardputer-Adv | G3 | G4 | G6 | G40 | G14 | G39 | G5 |
|---|---|---|---|---|---|---|---|
| Cap LoRa868 (LoRa) | RST | IRQ | BUSY | SCK | MOSI | MISO | NSS |

| Cardputer-Adv | G13 | G15 |
|---|---|---|
| Cap LoRa868 (GPS) | GPS-RX | GPS-TX |

# Model Size

- Cap LoRa868 Model Size PDF




# Datasheets

- SX1262
- ATGM336H-6N
- CASIC Multimode Satellite Navigation Receiver Protocol Specification

# Softwares

## Arduino

- Cap LoRa868 Arduino Quick Start
- Cap LoRa868 Arduino LoRa Driver Library
- Cap LoRa868 Arduino GPS Driver Library

## UiFlow2

- Cap LoRa868 UiFlow2 Docs

# Video

- Cap LoRa868 Product Introduction

U201-Cap-LoRa868-en.mp4